Responsibility as a Multi-Layer Signal: A Unified Computational Framework for Accountability Convergence in Regulatory Investigations

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Abstract Responsibility in regulatory investigations is not a single label — it is a multi-layered phenomenon produced through language, relational structure, and investigative process simultaneously. I present a proof-of-concept computational framework that operationalizes accountability convergence across three representational layers: process-level premature convergence detection (1), structural accountability graph extraction (3), and linguistic responsibility attribution prediction (4). These three frameworks are integrated into a composite Responsibility Convergence Index (RCI) (5) through PCA compression and Ridge Regression weight learning. Applied to an FDA Warning Letter corpus (N=50, 160 violations, 2016–2025, 21 CFR 210/211), the unified framework produces five cross-framework findings: (1) process-structural signals dominate accountability convergence in enforcement documents ($w = 0.995$); (2) linguistic attribution is the most individually distinct signal (Individual AUC = 0.940) but contributes minimally to composite convergence; (3) WL2018-05 is identified as the extreme convergence case across all four frameworks independently; (4) FDA Warning Letters exhibit structural pre-convergence in which Firm attribution dominates (99%) due to enforcement document logic rather than investigative reasoning; and (5) accountability convergence is weight-invariant at its extremes, with WL2018-05 ranking first across all weight schemes. Together, these findings advance a computational approach to accountability that moves beyond outcome assessment toward process quality. Given the use of theoretically-motivated synthetic labels, this framework is intended as a proof-of-concept operationalization rather than a validated detection system.

Keywords: computational accountability, responsibility convergence, multi-layer signal, regulatory NLP, FDA Warning Letters, premature convergence, accountability graphs, attribution prediction, composite index

Introduction

Accountability in regulated environments is not merely a legal outcome — it is produced through sequential interpretive decisions that unfold across multiple representational layers simultaneously. When an investigator writes "failure to establish adequate written procedures," they are constructing Firm-level accountability through linguistic choice (4). When that construction is represented as a directed graph, Firm emerges as the dominant hub with in-degree centrality of 0.444 (3). When the investigative process that produced that construction is examined for premature closure, 24% of FDA Warning Letter cases show elevated convergence risk (1). Each of these observations captures a different layer of the same underlying phenomenon: accountability convergence.

These three observations are not independent findings — they are converging measurements of the same latent construct. Prior computational work has examined accountability through single-signal lenses: as a fairness outcome (6), a transparency property (7), or a convergence process (8). No prior framework integrates linguistic, structural, and process-level signals into a unified measure of accountability convergence, nor examines what the relationship between these signals reveals about the structure of regulatory enforcement documents.

I address this gap by presenting a proof-of-concept computational framework that treats responsibility as a multi-layer signal and integrates four independently validated computational frameworks into a single composite index. The framework is applied to FDA Warning Letters (N=50, 160 violations, 2016–2025, 21 CFR 210/211) — a corpus that represents the final output of regulatory enforcement investigations in finished pharmaceutical manufacturing. The resulting cross-framework analysis reveals not only a quantitative measure of accountability convergence per case, but a structural property of regulatory enforcement documents themselves: they pre-converge accountability before investigative analysis begins.

This paper makes four contributions:

- I present a unified computational framework for accountability convergence that integrates process, structural, and linguistic signals into a single composite index with empirically derived weights.

- I demonstrate cross-framework convergent validity: WL2018-05 is identified as the extreme convergence case independently across all four frameworks operating on different representational levels.
- I provide empirical evidence that process-structural signals dominate accountability convergence in FDA Warning Letters ($w = 0.995$), while linguistic signals capture distinct within-case variation invisible to structural analysis.
- I characterize the structural pre-convergence of regulatory enforcement documents and derive design implications for accountability-aware AI systems.

The remainder of this paper is organized as follows. Section reviews related work across computational accountability, graph analysis, linguistic construction of responsibility, and composite metrics. Section presents the four-framework architecture. Section reports cross-framework analysis and findings. Section derives design implications. Section discusses limitations. Section concludes.

Related Work

Accountability in Computational Systems

Accountability has been studied extensively in public administration (8) and AI governance (6), but computational approaches to accountability as a measurable, multi-layered construct remain limited. Bovens (8) defines accountability as a social relationship in which actors explain and justify conduct to a forum, which may impose consequences. This definition is process-oriented — accountability is constituted through an interaction, not merely assigned as an outcome. The process orientation is consequential: it implies that accountability can be assessed not only by examining the final attribution, but by examining the investigative process that produced it and the document structures through which it was communicated. I extend this framing computationally, treating the investigative process, the relational structure of enforcement documents, and the linguistic choices within them as jointly constitutive of accountability convergence.

Prior computational work has focused on outcome fairness (6), model transparency (7), and algorithmic auditing. These approaches share a common limitation: they assess accountability after it has been assigned, without examining the process that produced the assignment. I address this limitation by treating accountability convergence as a property of the investigative and documentary process, detectable before and independent of outcome correctness.

Premature Closure in Investigative Reasoning

Premature closure — the tendency to settle on an explanation before exhausting alternatives — has been documented in medical diagnosis (9), legal reasoning, and organizational investigation. Croskerry (9) identifies premature closure as the most common cognitive error in clinical decision making, characterizing it as the failure to consider alternative diagnoses once an initial hypothesis has been accepted. The mechanism generalizes: in any investigative domain, early commitment to a single explanation forecloses the consideration of alternatives that may be equally or more plausible. Existing work relies on qualitative analysis or expert judgment rather than computational detection. I operationalize premature closure as a detectable signal in regulatory investigation documents, using structural and linguistic features to classify convergence risk without requiring expert annotation (1).

Graph-Based Accountability Analysis

Network analysis provides formal methods for examining relational accountability structures (10). Centrality measures — including in-degree, betweenness, and PageRank — quantify the structural importance of nodes within a network. Knowledge graph construction and relation extraction (11) have been applied to biomedical, legal, and scientific domains, but not to regulatory enforcement documents as accountability structures. I extend this methodology to accountability graphs, treating regulatory actors and CFR sections as nodes and enforcement relationships as typed directed edges, enabling structural analysis of accountability convergence at the case level (3).

Linguistic Construction of Accountability

Accountability in regulatory documents is not merely recorded — it is linguistically constructed through word choice, syntactic framing, and discourse patterns. NLP-based classification has been applied to clinical notes (12), legal documents, and compliance text, demonstrating that institutional language encodes social judgments in detectable lexical patterns. LIME (13) provides local, model-agnostic explanations for text classifiers, enabling identification of the specific tokens that drive accountability class predictions. I extend this work to responsibility attribution, treating the linguistic construction of accountability as a classification target and the resulting token-level explanations as evidence of the mechanisms through which responsibility is constituted in regulatory text (4).

Composite Metrics and Signal Integration

Composite indices aggregate multi-dimensional constructs into single scores, enabling comparison across cases and identification of extreme values (14). The OECD Handbook on Constructing Composite Indicators (14) identifies weight specification, multicollinearity, and normalization as the primary methodological challenges in composite index construction. Multi-layer signal fusion methods combine heterogeneous data sources into unified representations (15), with PCA and learned weighting as standard approaches to resolving signal redundancy. I apply these methods to accountability convergence, using VIF analysis to detect multicollinearity, PCA to compress correlated signals, and RidgeCV to learn empirically grounded weights (5).

The Four-Framework Architecture

Framework 1: Premature Convergence Detection

Research question. Can premature accountability convergence be detected as a process-quality signal in regulatory investigation documents, distinct from outcome correctness?

Method. I operationalize premature accountability convergence (PAC) through six features derived from reasoning logs or investigative documents: explanation count, diversity score

(Shannon entropy over explanation categories), commitment level, uncertainty marker, closure language, and alternative considered. Labels are assigned probabilistically via a risk score:

$$r = w_1(3 - e) + w_2 c + w_3(1 - u) + w_4 l \tag{1}$$

where e, c, u, l denote explanation count, commitment level, uncertainty marker, and closure language respectively, and $w_1 = w_2 = w_3 = w_4 = 0.25$. Three classifiers (Logistic Regression, Random Forest, SVM) are evaluated via 5-fold cross-validation on a synthetic dataset (n=500) and validated against the FDA Warning Letter corpus.

Results. All three classifiers achieve F1 scores of 0.901–0.913 and AUC above 0.97, indicating that premature convergence is robustly detectable across model architectures. Ablation analysis identifies commitment level as the strongest predictor (ΔF1 = 0.060). SHAP analysis reveals that explanation count drives case-level variation, suggesting a two-signal model: commitment as a necessary condition, explanation breadth as the discriminating factor. Applied to FDA Warning Letters, 12/50 cases (24%) exceed the convergence risk threshold, with WL2018-05 exhibiting the maximum risk score of 1.0 — a single violation with zero explanatory diversity. Figure 1 shows the resulting per-case RCI distribution, in which WL2018-05 is the sole outlier above mean + 1 SD.

Framework 2: Accountability Graph Extraction

Research question. What is the relational structure of accountability in regulatory enforcement documents, and how does structural convergence relate to investigative closure?

Method. I define three accountability relation types derived from the structure of FDA Warning Letters: VIOLATES (responsibility subject $\rightarrow$ CFR section), INVOLVES (violation instance $\rightarrow$ violation text), and ATTRIBUTED_TO (violation instance $\rightarrow$ responsibility subject). For each violation record, I extract a unique violation node, a responsibility subject node, a CFR node, and a violation text node. A composite accountability openness score combines violation count, CFR diversity, and text diversity:

$$o = w_1 \cdot \tilde{n}_v + w_2 \cdot \tilde{n}_c + w_3 \cdot \tilde{d}_t \tag{2}$$

where $w_1 = 0.3$, $w_2 = 0.4$, $w_3 = 0.3$.

Results. The framework extracts 320 accountability triples across 50 FDA Warning Letter cases, constructing a directed graph with 359 nodes and 363 edges across 68 unique CFR sections. Graph analysis reveals star-topology accountability structure: Firm exhibits in-degree centrality of 0.444, receiving 44% of all incoming accountability edges, with zero clustering coefficient across all nodes. The accountability openness score exhibits significant negative correlation with PAC convergence risk (Pearson $r = -0.431$, $p = 0.0018$; Spearman $\rho = -0.742$, $p < 0.001$), providing the first cross-framework validation between structural and process-level signals. Figure 4 visualizes this relationship, with WL2018-05 as an extreme outlier in both dimensions. K-means clustering ($k = 3$) identifies three structural case types: Open (n=16, openness ≥ 0.674), Standard (n=33, openness 0.373–0.665), and Extreme convergence (n=1, WL2018-05, openness = 0.039).

Table 1: Accountability Graph Statistics (N=50 FDA Warning Letters)

Metric	**Value**
Total nodes	359
Total edges	363
Unique CFR sections	68
Firm in-degree centrality	0.444
Firm betweenness centrality	0.052
Clustering coefficient	0.000
Spearman ρ (openness vs PAC risk)	-0.742^{***}

$^{***}p < 0.001$

Framework 3: Responsibility Attribution Prediction

Research question. Can responsibility attribution class (Firm / Individual / System) be predicted from violation text alone, and what linguistic patterns drive each class?

Method. I define responsibility attribution as a three-class text classification task. TF-IDF vectorization (unigram + bigram, max 200 features, sublinear TF normalization) represents violation text. Three classifiers are evaluated via Leave-One-Out cross-validation on a manually

annotated corpus (n=45, 15 per class). LIME analysis reveals class-specific linguistic markers. The trained model is applied to FDA Warning Letters for external validation.

Results. Logistic Regression achieves the highest macro F1 (0.725), outperforming Random Forest (0.712) and SVM (0.556). Per-class ROC analysis (Figure 3) shows that Individual-level attribution is most linguistically distinct (AUC = 0.940), driven by agent-noun signals (*operator*, *analyst*, *employee*). System-level attribution exhibits the lowest discriminability (AUC = 0.764), reflecting genuine linguistic overlap with Firm-level language: infrastructure failures are often described using organizational framing (*inadequate controls*, *failure to establish*). Applied to FDA Warning Letters, the model reassigns 49/160 Firm-labeled violations to System (30.6%), with uniformly low prediction confidence (0.35–0.65), indicating linguistic ambiguity between classes in real regulatory text.

Table 2: Responsibility Attribution Classification Results (LOO-CV)

Model	Macro F1	Best class	Best AUC
Logistic Regression	0.725	Individual (0.85)	0.940
Random Forest	0.712	Individual (0.83)	0.931
SVM	0.556	Individual (0.72)	0.871

Framework 4: Responsibility Distribution Metric

Research question. Can process, structural, and linguistic signals be integrated into a unified accountability convergence index, and what does the learned weight structure reveal about regulatory enforcement documents?

Method. I define the Responsibility Convergence Index (RCI) as:

$$RCI = w_{ps} \cdot \tilde{z}_{ps} + w_{\ell} \cdot \tilde{z}_{\ell} \tag{3}$$

where $\tilde{z}_{ps}$ is the MinMax-normalized process-structural score (PCA compression of PAC risk and Centrality Gini; PC1 = 98.6% variance explained) and $\tilde{z}_{\ell}$ is the MinMax-normalized linguistic convergence score ($1 - H_{norm}$, where H_{norm} is normalized Shannon entropy of predicted attribution class distributions). Weights are learned via RidgeCV (5-fold CV, $\alpha \in$

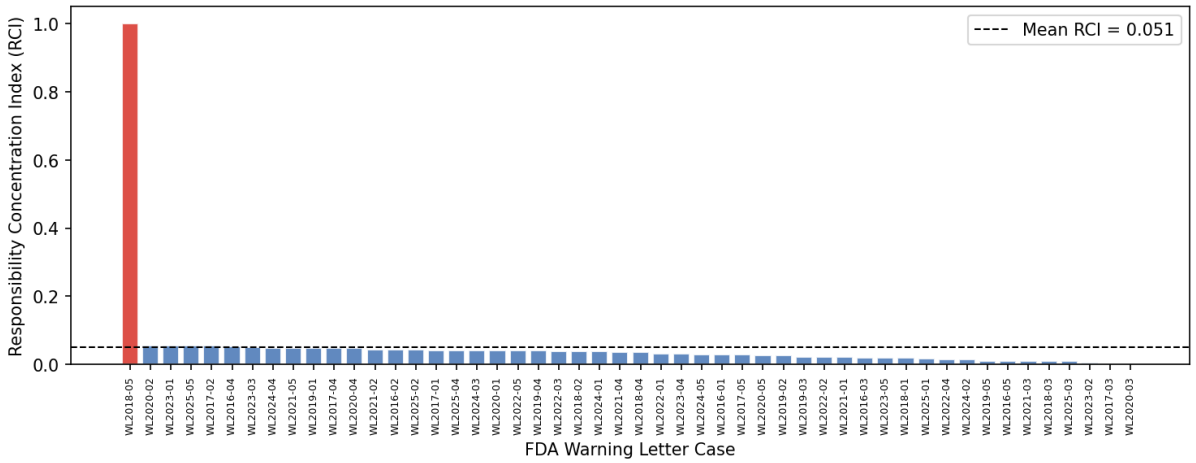


Figure 1: RCI scores across 50 FDA Warning Letter cases, sorted descending. Red = above mean + 1 SD. WL2018-05 (RCI = 1.000) is the sole extreme case, identified consistently across all four frameworks.

$\{0.01, 0.1, 1, 10, 100\}$). VIF analysis confirms input independence post-compression (VIF = 1.03).

Results. The optimal regularization strength is $\alpha = 100$, with $R^2 = 0.005$ (linguistic $\rightarrow$ process-structural), yielding learned weights $w_{ps} = 0.995$, $w_\ell = 0.005$. Figure 1 shows RCI scores across all 50 cases. RCI mean = 0.051, SD = 0.138, with WL2018-05 at RCI = 1.000 (rank 1/50). Sensitivity analysis across four weight schemes (Figure 2) confirms rank robustness ($\rho > 0.97$) among Equal, PS-dominant, and Attr-dominant schemes, with the Learned scheme producing a qualitatively distinct ranking ($\rho = 0.31$–0.43) that itself constitutes an empirical finding about structural signal dominance.

Table 3: RCI Weight Learning and Sensitivity Analysis

Weight scheme	w_{ps}	w_ℓ
Learned (Ridge, $\alpha = 100$)	0.995	0.005
Equal	0.500	0.500
PS-dominant	0.750	0.250
Attr-dominant	0.250	0.750
ρ (Equal vs PS-dominant) = 0.982		
ρ (Equal vs Attr-dominant) = 1.000		
ρ (Learned vs Equal) = 0.315		

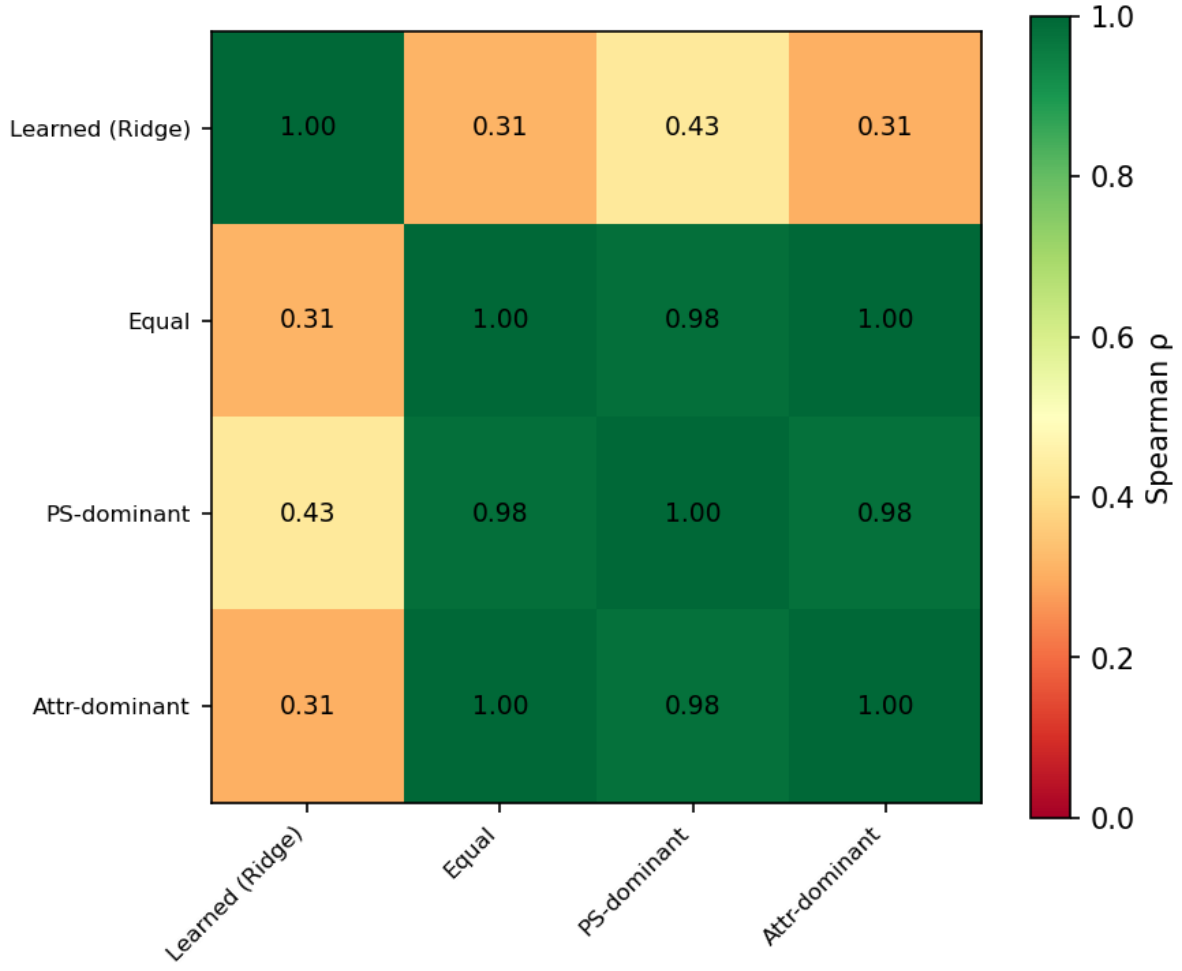


Figure 2: Spearman rank correlation across four weight schemes. Equal, PS-dominant, and Attr-dominant schemes produce near-identical rankings ($\rho > 0.97$). The Learned (Ridge) scheme is qualitatively distinct, reflecting near-complete process-structural dominance.

Cross-Framework Analysis

Convergent Validity: WL2018-05

The most consequential cross-framework finding is the independent identification of WL2018-05 as the extreme convergence case across all four frameworks. Table 4 reports WL2018-05 scores across all frameworks.

WL2018-05 is a 2018 FDA Warning Letter with a single violation, zero explanatory diversity, and pure Firm attribution. Its convergent identification across four independently developed frameworks — each operating on a different representational level, using different methods, and validated independently — constitutes strong cross-method evidence of structural accountability convergence. Crucially, no framework was designed with knowledge of the others' outputs during development; the convergence is emergent rather than engineered.

Signal Independence and Complementarity

A key finding of the cross-framework analysis is that the three signal layers are empirically independent while measuring complementary aspects of accountability convergence. VIF analysis post-PCA compression confirms independence (VIF = 1.03). The low R^2 between linguistic and process-structural signals ($R^2 = 0.005$) is not a failure of integration — it is evidence that the two layers capture distinct variance in the corpus.

This independence has a direct interpretive consequence: the near-zero learned weight for the linguistic layer ($w_\ell = 0.005$) does not indicate that linguistic signals are uninformative. Framework 3 demonstrates that linguistic signals achieve AUC = 0.940 for Individual attribution and detect systematic Firm-System misattribution in 30.6% of FDA violations. Rather, the low weight indicates that in FDA Warning Letters specifically, the variance explained by linguistic signals is not the same variance that drives composite convergence scores. This is itself a finding about the corpus: structural signals dominate because enforcement document architecture pre-determines attribution patterns in ways that linguistic choices cannot override.

Structural Pre-Convergence as a Corpus Property

Across all four frameworks, FDA Warning Letters exhibit a consistent pattern I term *structural pre-convergence*: accountability is concentrated toward Firm attribution before any investigative analysis begins, as a consequence of enforcement document architecture rather than investigative reasoning. This pattern is consistent with Bovens' (8) process-oriented definition of accountability — the enforcement document does not merely record the outcome of an accountability interaction; it constitutes one, pre-determining the forum (the regulator), the actor (the firm), and the obligation (corrective action) through document structure before any investigative language is written.

The computational evidence for structural pre-convergence is consistent across all four frameworks. Framework 1 finds that 24% of cases show elevated PAC risk, with WL2018-05 having a single violation and zero explanatory diversity — a document structure that forecloses alternative attributions by design. Framework 2 finds that 159/160 actual responsibility labels are Firm, producing a star-topology graph with zero clustering coefficient — a structural signature of pre-convergence visible in Figure 4. Framework 3 finds that the model reassigns 30.6% of Firm-labeled violations to System, with uniformly low prediction confidence (0.35–0.65), indicating that the linguistic content of many Firm-labeled violations is systematically ambiguous with respect to the assigned label. Framework 4 finds that RCI scores cluster near zero for 49/50 cases (mean = 0.051), with WL2018-05 as a clear outlier — a distribution consistent with a corpus in which most cases share a similar structural pre-convergence baseline.

Together, these findings characterize structural pre-convergence as a computationally detectable property of regulatory enforcement documents, not merely a descriptive observation about label distributions.

Table 4: WL2018-05 Scores Across All Four Frameworks

Framework	Metric	Value
F1 (Process)	Convergence risk	1.000 (max)
F2 (Structural)	Openness score	0.039 (min)
F3 (Linguistic)	Attribution entropy	0.000 (min)
F4 (Composite)	RCI	1.000 (rank 1/50)

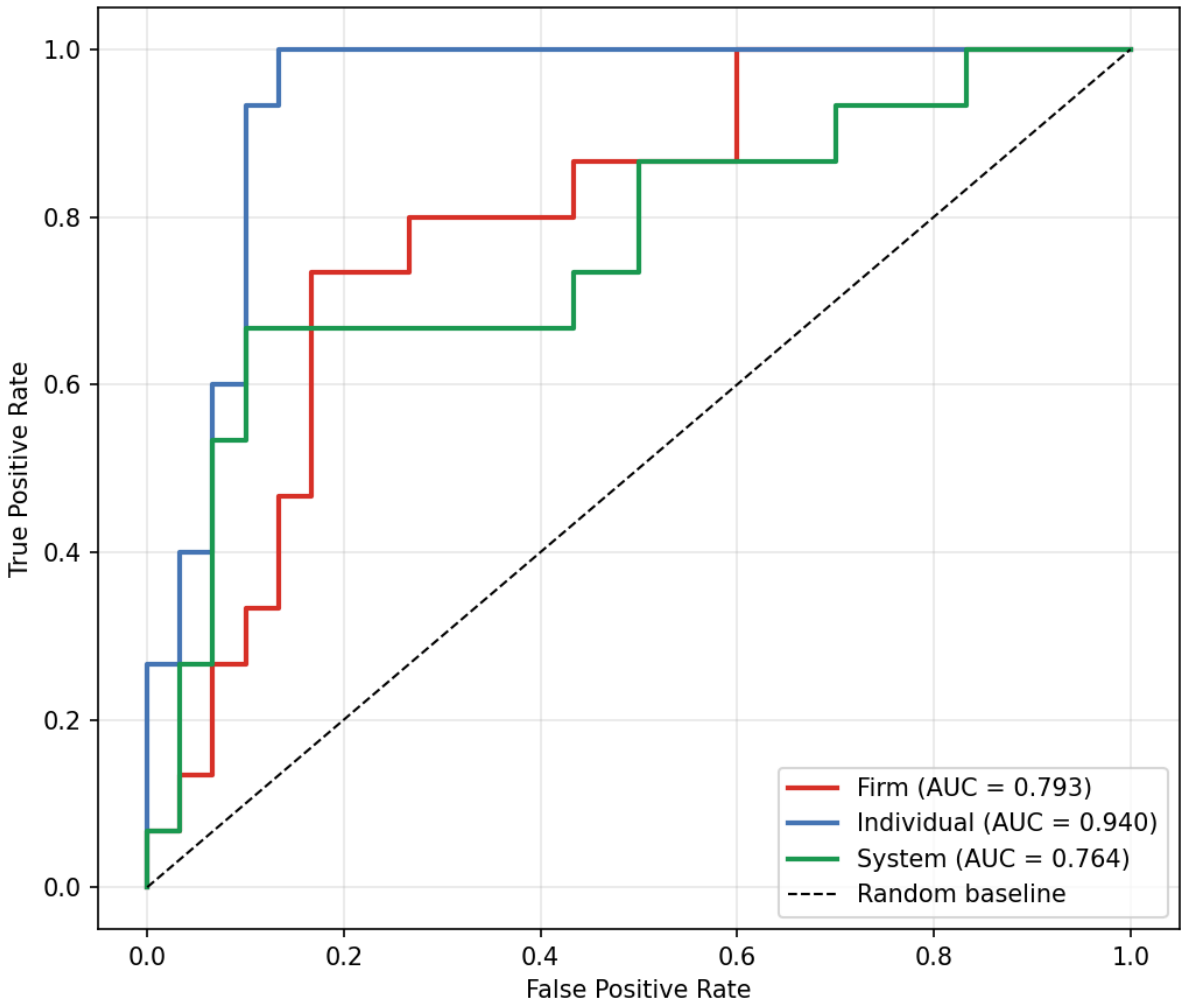


Figure 3: Per-class ROC curves for responsibility attribution prediction (LOO-CV). Individual achieves AUC = 0.940; System achieves AUC = 0.764, reflecting genuine linguistic overlap with Firm-level framing.

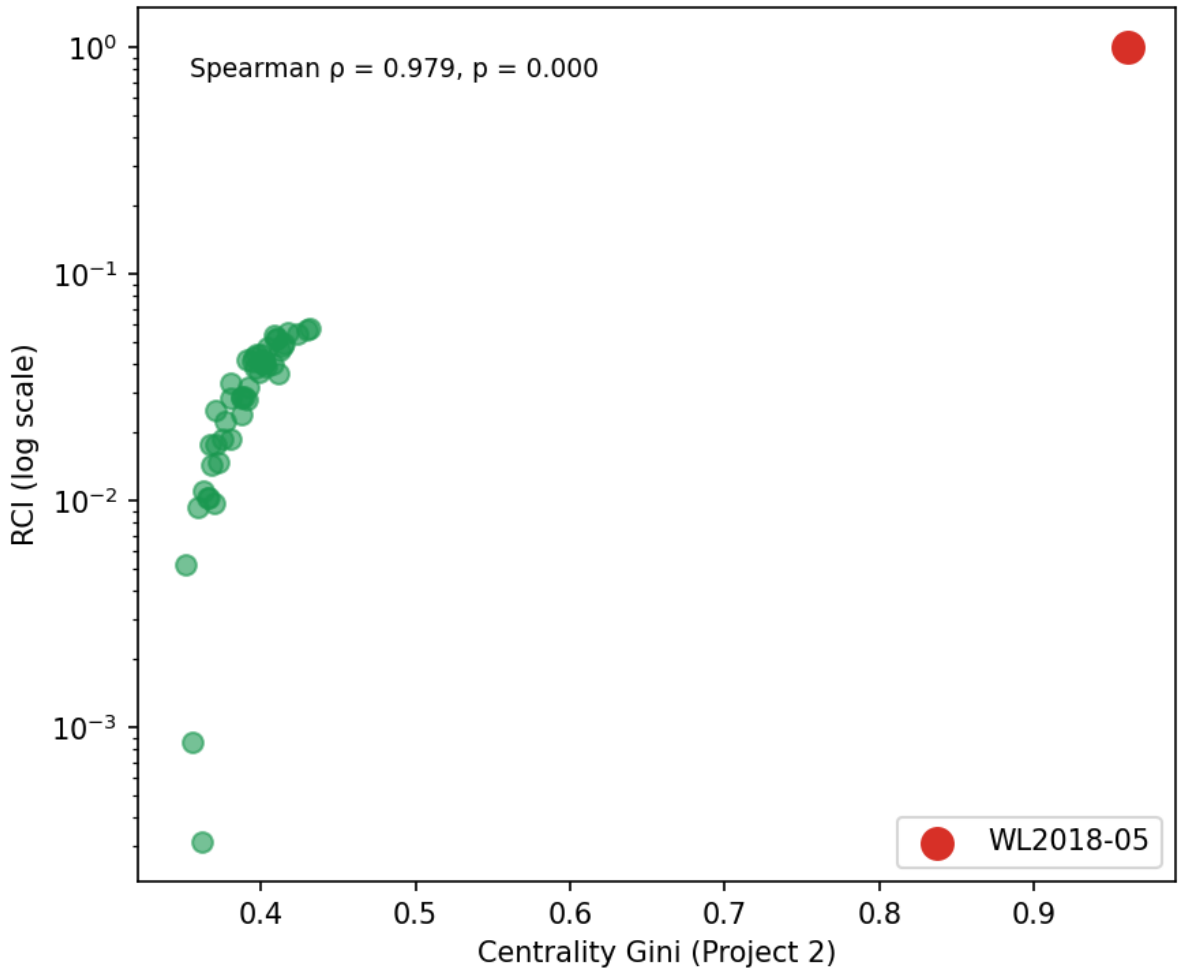


Figure 4: Cross-framework validation: RCI vs. Centrality Gini (log scale y-axis, Spearman $\rho = 0.979$, $p < 0.001$). WL2018-05 is an extreme outlier in both dimensions, confirming weight-invariant detection of maximum accountability convergence.

Implications for Accountability-Aware AI Design

Process Quality over Outcome Accuracy

The unified framework advances a shift from outcome-oriented to process-oriented accountability assessment. Framework 1 demonstrates that premature convergence — not outcome error — is the primary accountability risk in investigative reasoning: a conclusion can be correct and still reflect premature closure if alternatives were not adequately explored. Accountability-aware AI systems should therefore flag investigative closure before alternatives have been exhausted, rather than verify the correctness of final conclusions. This reframes the design problem: the goal is not to automate accountability assignment, but to keep the comparison alive.

Structural Signals Must Supplement Linguistic Ones

Framework 3 demonstrates that linguistic classification alone achieves macro F1 = 0.725 — sufficient for detection but imperfect, with systematic ambiguity at the System-Firm boundary (AUC = 0.764). Framework 4 shows that structural signals dominate composite convergence ($w_{ps} = 0.995$). Accountability-aware AI systems that rely solely on text classification will systematically underestimate structural convergence and conflate System-level and Firm-level failures — precisely the conflation that Framework 3 identifies in 30.6% of FDA violations. Graph-level structural features are therefore not optional enhancements but necessary components of robust accountability assessment.

Enforcement Document Structure Is Not a Neutral Input

Framework 2 reveals that FDA Warning Letter accountability graphs exhibit star topology with Firm as the sole hub — a structural property of enforcement document architecture. Framework 3 shows that this architecture systematically masks infrastructure failures under organizational accountability. AI systems trained on regulatory documents without accounting for this structural pre-convergence will inherit and amplify it (2): they will learn to assign Firm attribution not because Firm attribution is most accurate, but because enforcement document structure makes it the overwhelmingly dominant label. Responsible development of accountability AI in regulatory

contexts requires explicit modeling of the pre-convergence baseline as a corpus artifact, not a ground truth.

Limitations

Corpus scope. All four frameworks are applied to FDA Warning Letters (21 CFR 210/211), a corpus with extreme Firm attribution dominance (159/160 actual labels). The star-topology graph structure, near-zero RCI distribution, and structural pre-convergence findings are properties of this specific corpus. Generalizability to other regulatory domains — healthcare incident reports, aviation safety investigations, financial enforcement actions — requires validation against domain-specific corpora with greater responsibility subject diversity.

Synthetic data components. Frameworks 1 and 4 rely on synthetic data: probabilistic labeling for PAC detection (n=500 synthetic cases) and reconstructed violation texts for attribution entropy computation. Because synthetic labels are derived from the same theoretical framework being operationalized, the classification results constitute an illustration of the framework's internal consistency rather than independent empirical validation. The synthetic-to-real validation in Framework 1 demonstrates ecological validity (diversity score $\Delta = 0.017$ between synthetic and FDA Warning Letter distributions), but integration of synthetic signals amplifies rather than cancels modeling assumptions. Human annotation of in-process investigative reasoning logs remains a necessary step for ground-truth validation.

Document type limitation. FDA Warning Letters represent final regulatory outputs rather than in-process investigative reasoning logs. Framework 2's star-topology finding and Framework 1's convergence risk scores reflect the structure of enforcement documents, not necessarily the investigative processes that preceded them. Direct access to CAPA logs, root cause analysis records, and deviation investigation reports would enable observation of accountability convergence as it unfolds.

Linguistic model scope. Framework 3's annotated corpus is small (n=45) and domain-specific. TF-IDF features do not capture syntactic structure, discourse context, or cross-sentence dependencies. The System-Firm boundary ambiguity (AUC = 0.764) likely reflects not only

genuine linguistic overlap but also the limitations of surface-level feature representations. Transformer-based models with dependency parsing may resolve ambiguities that TF-IDF cannot.

Weight generalizability. The RCI weight structure ($w_{ps} = 0.995$) reflects corpus-specific dominance of structural signals in FDA Warning Letters. In corpora with greater responsibility subject diversity — where Individual and System attributions appear with meaningful frequency — linguistic signals may contribute more substantially to composite convergence, and the learned weights would differ accordingly.

Integration approximation. Cross-framework integration relies on per-case score reconstruction rather than a unified end-to-end pipeline with shared feature representations. Each framework was developed independently with its own data generation and modeling assumptions. A tightly integrated pipeline with shared representations and joint training would likely yield different weight structures and may reveal signal interactions not visible in the current sequential integration.

Conclusion

I presented a unified computational framework that operationalizes responsibility as a multi-layer signal — detectable at the linguistic, structural, and process level simultaneously. The framework integrates four independently validated computational frameworks into a composite Responsibility Convergence Index through PCA compression and Ridge Regression weight learning, applied to FDA Warning Letters (N=50, 160 violations, 2016–2025).

The cross-framework analysis produces five findings that collectively characterize accountability convergence in regulatory enforcement documents. First, process-structural signals dominate composite convergence ($w_{ps} = 0.995$), reflecting the dominance of document architecture over linguistic choice in FDA Warning Letters. Second, linguistic signals are empirically independent of structural signals (VIF = 1.03) and capture distinct within-case variation — particularly at the System-Firm boundary — that structural analysis cannot observe. Third, WL2018-05 exhibits convergent validity as the extreme convergence case across all four frame-

works independently, providing cross-method validation of structural accountability convergence. Fourth, FDA Warning Letters exhibit structural pre-convergence: accountability is concentrated toward Firm attribution by enforcement document architecture before investigative analysis begins — a pattern consistent with Bovens' (8) process-oriented definition of accountability, in which document structure itself constitutes an accountability interaction. Fifth, extreme accountability convergence is weight-invariant: WL2018-05 ranks first across all weight schemes, suggesting that maximum convergence produces a signal strong enough to dominate regardless of how layers are weighted.

Future work will extend this framework to richer investigative corpora with direct access to in-process reasoning logs, incorporate transformer-based linguistic representations, develop multi-domain validation across healthcare, aviation, and financial regulatory environments, and pursue joint training of the four-framework pipeline to enable shared feature representations and cross-layer signal interaction modeling.

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